

Program : Diploma in Instrumentation Engineering	
Course Code : 5083A	Course Title: Industrial Electronics and Drives
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week : 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To give basic principles of different power semiconductor diodes, its working, characteristics and applications.
- To get a basic knowledge on the working of DC and AC motors.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics II	2
Basic knowledge of semiconductors		Electronic Instrumentation	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the working and application of power semiconductor devices	15	Understanding
CO2	Explain the working and application of SCR in switching circuits	15	Understanding
CO3	Explain the working principle of DC & AC Motors	14	Understanding
CO4	Explain single phase and three phase dual converters, choppers and electric drives.	14	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3					2	
CO3	3						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the working and application of power semiconductor devices.		
M1.01	Define semiconductor and describe the working of power diodes	3	Understanding
M1.02	Describe the characteristics of SCR	4	Understanding
M1.03	Describe the working of TRIAC and DIAC	4	Understanding
M1.04	Describe different protection methods of SCR	4	Understanding
Contents: Power Semiconductor Devices - Power diode - Structure and VI characteristics - SCR - Construction and features - operation - transistor analogy - characteristics - specifications - holding current - latching current - gate current - turn on time-turn off time - TRIAC - structure, working, static characteristics - DIAC - structure, working, static characteristics - Protection methods - Snubber circuit.			
CO2	Explain the working and application of SCR in switching circuits		
M2.01	Describe triggering methods of SCR	4	Understanding
M2.02	Explain commutation techniques of SCR	4	Understanding
M2.03	Describe operation of inverters	4	Understanding
M2.04	Describe static switch and PWM	3	Understanding
	Series Test - I	1	
Contents: Converters - Triggering and Commutation of SCR - Triggering of SCR - different methods of turn on - R triggering - RC triggering - triggering using UJT relaxation oscillator -			

Commutation of SCRs - Line commutation and forced commutation -Operation of single-phase and three phase inverters-TRIAC light dimming circuit - Static switches-Pulse width modulation (PWM)- PWM Chip-SG352

CO3	Explain the working principle of DC & AC Motors		
M3.01	Describe the working of DC motors	2	Understanding
M3.02	Describe the classification of DC motors	4	Understanding
M3.03	Describe the working of Induction motors	4	Understanding
M3.04	Explain the working of servo motors and stepper motors	4	Understanding

Contents:

DC & AC Motors: Principle of operations of DC motor - Fleming's left hand rule - back emf - torque equation - load characteristics of shunt and series motors - single phase induction motor - Three phase induction motor-Universal motor - application of single phase Induction motor - methods of speed control of induction motors - variable voltage - variable frequency - stepper motors - principle - applications - servo motors - tachogenerators - principle-operation - application - Soft start of AC and DC motors

CO4	Explain single phase and three phase dual converters		
M4.01	Explain the working of converters.	5	Understanding
M4.02	Describe the working of choppers	5	Understanding
M4.03	Compare AC and DC Drives	4	Understanding
	Series Test - II	1	

Contents:

Control Drives: Dual converters - Single phase and three phase dual converters - principle and operation waveforms - Application - Cycloconverter for single phase - applications. Choppers - Methods of output control - principle of step up and step down chopper - Jone's chopper principle and working - waveforms. AC chopper - application of choppers - Definition of drives - comparison between AC and DC drive - Requirement of variable speed drive - Speed control of DC drives.

Text / Reference:

T/R	Book Title/Author
R1	B.L.Theraja, A Text Book of Electrical Eng volume 2, S Chand Publishers
R2	P. S. Bimbhra, Power Electronics, Khanna Publishers
R3	Biswanath Paul, Industrial Electronics and Control ,PHI Publishers
R4	Bimal.K.Bose, Modern Power Electronics and AC Drives, PHI Publishers
R5	Joseph Vithayathil, Power Electronics Principles and Applications, McGraw Hill Publishers
R6	S.K.Bhattacharya, S.chaterjee , Industrial Electronics and Control, Technical Education Series

Online Resources:

Sl. No	Website Link
1	https://www.tutorialspoint.com/power_electronics/index.htm
2	https://www.electronics-tutorials.ws/power/thyristor.html
3	https://nptel.ac.in/courses/108108077/
4	https://www.elprocus.com/electric-drive-types-block-diagram-classification/
5	https://automationforum.in/t/what-are-electrical-drives/4841